

PROBLEM SET 5

ECON301—ECONOMETRICS I

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Note: in the lecture notes and videos, I usually denote vectors by underlining them, for example $\underline{x}$, and I denote matrices in capital letters. When typesetting it, I use **bold notation and upright letters** to denote vectors and matrices, i.e., $\mathbf{x}$ and $\mathbf{M}$ for example.

1. In a study to evaluate the effects of capital (K) and labor (L) on production, a sample of 75 agricultural businesses was used. The model fitted was

$$E^*(\ln(Prod) | \ln(L), \ln(K)) = \gamma_0 + \gamma_1 \ln(L) + \gamma_2 \ln(K) \quad (1)$$

The parameters γ_1 and γ_2 are called the *output elasticities* of the labor and capital inputs respectively. Suppose that Assumption 1M is reasonable for this data since it comes from a survey. Assumption 2M is also OK since the routine to fit the model did it without sign of trouble, i.e., the regression matrix was of full rank. The results of the estimation procedure are given in tables 1 and 2.

Variable	Coefficient	Std. Error
Intercept	-3.3387546	2.4495134
$\ln(L)$	1.49883206	0.5398014
$\ln(K)$	0.48985174	0.102044

Table 1: Fitting results of model in (1)

$$\hat{\Omega} = \begin{pmatrix} 6.00011581 & -1.2605606 & 0.1121914 \\ -1.2605606 & 0.29138551 & -0.0384267 \\ 0.1121914 & -0.0384267 & 0.010413 \end{pmatrix}$$

Table 2: Estimated variance-covariance matrix from model (1)

- i. Is there evidence that the output elasticity of labor, γ_1 , is strictly positive? Use a 5% confidence interval. Compute the P -value also.
 - ii. Test if there is evidence that the output elasticity of labor is strictly greater than the output elasticity of capital. Use a 5% significance level. Compute the P -value also.
 - iii. Test if there is evidence that $\gamma_2 > 1$. Use a 5% level of significance. Compute the P -value also.
 - iv. Test if there is evidence that $\gamma_2 > 1.2$. Use a 5% level of significance. Compute the P -value also.
 - v. Test if there is evidence of *increasing returns to scale*, i.e. $\gamma_1 + \gamma_2 > 1$. Use a 5% level of significance. Compute the P -value also.
 - vi. Make the test of global significance of the model. Use a 5% level of significance. Compute the P -value also.
2. Let (X_i, Y_i) $i = 1, 2, \dots$ i.i.d. observations from a joint pdf in which both X and Y have finite variances. Let $E^*(Y|X) = \alpha + \beta X$ be the BLP of Y as a function of X . Let $(\hat{\beta}_0, \hat{\beta}_1)^\top$ be the LSE vector.
 - i. Using the fact that $\hat{\beta}_1 \rightarrow_P \beta$ when $n \rightarrow \infty$, show that $\hat{\beta}_0 \rightarrow_P \alpha$ when $n \rightarrow \infty$.
 - ii. Using the fact that $\hat{\beta}_1 \sim AN(\beta, \frac{\phi^2}{n})$, and that $\hat{\phi}^2 \rightarrow_P \phi^2$, where $\hat{\phi}^2$ is the consistent estimator of ϕ^2 , give the details of building an asymptotic pivot to make inference on β .
 - iii. Using (b) obtain an asymptotic $(1 - \alpha)\%$ confidence interval for β .
 - iv. Using (b) give the procedure to test $H_1 : \beta \neq \beta_0$ or $H_1 : \beta > \beta_0$ with $\alpha\%$ significance level.
 3. This problem is completely of didactic nature since the sample size is too small. Suppose (X_i, Y_i) $i = 1, \dots, 4$ are i.i.d. observations from a pdf $f(x, y)$. Suppose both variables have finite variance. The data points are given in table 3

i	X	Y
1	5	8
2	4	7.5
3	2	2.5
4	1	2

Table 3: Data for problem 3.

- Find the LSE.
 - Compute the residuals.
 - Compute $\hat{\phi}^2$.
 - Find a 95% confidence interval for β .
 - Is there evidence that $\beta \neq 0$? Use a 1% significance level.
4. Let ET = National terminal efficiency in elementary school measured as a percent, PAM = Average number of students per teacher at elementary level (total number of students divided by total number of teachers), $GEPR$ = Governmental national expenditure on elementary education, for a sample of 75 municipalities. The following model was fitted:

$$E^*(ET|PAM, GEPR) = \beta_0 + \beta_1 PAM + \beta_2 GEPR.$$

The results of this fit are shown below:

$$\hat{\beta} = \begin{pmatrix} 84.89061 \\ -1.003587 \\ 7.332466 \end{pmatrix}$$

$$\hat{\Omega} = \begin{pmatrix} 3.704246 & -0.119479 & 1.670767 \\ -0.119479 & 0.004506 & -0.117575 \\ 1.670767 & -0.117575 & 7.328432 \end{pmatrix}$$

- Make the global significance test of the model. Use a 5% level of significance. Compute the P -value.
 - Find a 95% confidence interval for β_1 . is the sign of β_1 consistent with common sense?
 - Test if there is evidence that $\beta_2 > 0$. Use a 5% level of significance. Compute the P -value.
5. Consider the data set in table 4 Fit the BLP $E^*(Y|X) = \beta_0 + \beta_1 X$.

i	X	Y
1	1	2
2	1	3
3	1	0
4	0	2
5	0	3

Table 4: Data for problem 5.

- Make the computations by hand to compute $\hat{\beta}$, $\hat{\epsilon}$ and $\hat{\Omega}$.
 - Make an R program to make the same computations and compare the results with those of (a).
6. Let $\{(X_i, Y_i)\}$ be a sequence of bivariate vectors such that assumptions 1M and 2M are true. Consider the notation of the proof of theorem 6.2.¹
- Let $W_i = (X_i - \mu_X)\epsilon_i$. Show that the sequence $\{W_i\}$ is i.i.d.
 - Show that $E(W_i) = 0$.
 - Show that $\text{Var}(W_i) = E(X_i^2 \epsilon_i^2) - 2\mu_X E(X_i \epsilon_i^2) + \mu_X^2 E(\epsilon_i^2)$.
 - Show that $E(\epsilon_i^2) = \text{Var}(Y_i) + \beta_1^2 \text{Var}(X_i) - 2\beta_1 \text{Cov}(X_i, Y_i)$.

¹Check slides 3–9 of lecture 6.

- v. Show that $\frac{1}{n} \sum_i \hat{\epsilon}_i^2$ is a consistent estimator of $E(\epsilon_i^2)$. Notice that the sequence $\{\hat{\epsilon}_i^2\}$ is not i.i.d. and hence the WLLN cannot be applied directly. *Hint: note that $\bar{\epsilon} = 0$ and hence $\frac{1}{n} \sum_i \hat{\epsilon}_i^2 = \frac{1}{n} \sum_i (\hat{\epsilon}_i - \bar{\epsilon})^2$ and substitute both $\hat{\epsilon}_i$ and $\bar{\epsilon}$ before squaring.*
- vi. Find a consistent estimator of $\mu_X^2 E(\epsilon_i^2)$.
- vii. Propose a consistent estimator of $E(X_i^2 \epsilon_i^2)$. It is not necessary to show its consistency.
- viii. Propose a consistent estimator of $2\mu_X E(X_i \epsilon_i^2)$. It is not necessary to show its consistency.
- ix. Put together (f), (g), and (h) and show that

$$\hat{\sigma}_W^2 = \frac{1}{n} \sum_i [(X_i - \bar{X}) \hat{\epsilon}_i]^2$$

is a consistent estimator of $\text{Var}(W_i)$.